

THE UNIVERSITY *of* EDINBURGH

Statistical Case Studies

Project 1, part 1: Uncertainty in river flow hydrographs for flood inundation modelling

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Word Count: XXXX

October 2026

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1 Executive Summary

Current flood risk models generally assume that the shape of a flood is fixed, varying only in its maximum river flow. This project investigates whether this approach captures the uncertainty in flood events on the River Don at Parkhill, by considering both the maximum flow and the time a flood takes to reach its peak.

Using ten years of river discharge data, we identified 44 independent extreme flow events, across which both quantities varied considerably, with median values of around 106 m^3/s and 44 hours respectively. Flood events were concentrated in autumn and winter, with autumn events typically taking longer to reach their peak, and we found only weak evidence of a relationship between maximum flow and time to peak, suggesting that larger floods do not necessarily develop more slowly.

These findings suggest that assuming a fixed hydrograph shape may overlook important uncertainty, so SEPA should consider incorporating variation in time to peak alongside maximum flow in future flood risk modelling guidance.

2 Data and Exploratory Analysis

2.1 Data Description and Quality

Table 1: Data used in this analysis, and the quality codes assigned to each reading

Data description	
Station	Don at Parkhill
Source	SEPA time series data service
Measurement	River discharge (m^3/s)
Catchment area	1,273 km^2
Resolution	15 minutes
Period	01 October 2016 – 30 September 2026
Readings	350,592
Quality codes	
Missing readings	0
50 – Good (G)	335,647 readings (95.74%), retained
100 – Estimated (E)	14,915 readings (4.25%), retained
140 – Provisional (Prov)	0 readings (0.00%), excluded
150 – Suspect (S)	0 readings (0.00%), excluded
200 – Unchecked (imported from legacy database) (V)	0 readings (0.00%), excluded
254 – Unchecked (U)	30 readings (0.01%), excluded

The subsequent analysis uses discharge readings (m^3/s) from the River Don at Parkhill velocity-area gauging station, covering 10 water years. Parkhill, the lowest gauging station on the Don, applies a single peak flow rating curve to all measurements.

Each reading is assigned a SEPA quality code (Table 1).

- Estimated ($n = 14,915$): Retained. SEPA defines these as acceptable rating curve extrapolations where the river level exceeds the 3.7 m gauging limit, at which Good readings are capped (Figure 1), so they contain the largest flows and preserve the extremes.
- Unchecked ($n = 162$): Excluded. These are restricted to recent entries (29 September 2026, 16:15 onward) and in sporadic 2017–2018 bursts of at most 45 minutes, and only

two coincided with extreme flows, so removing them leaves a fully verified dataset with negligible impact.

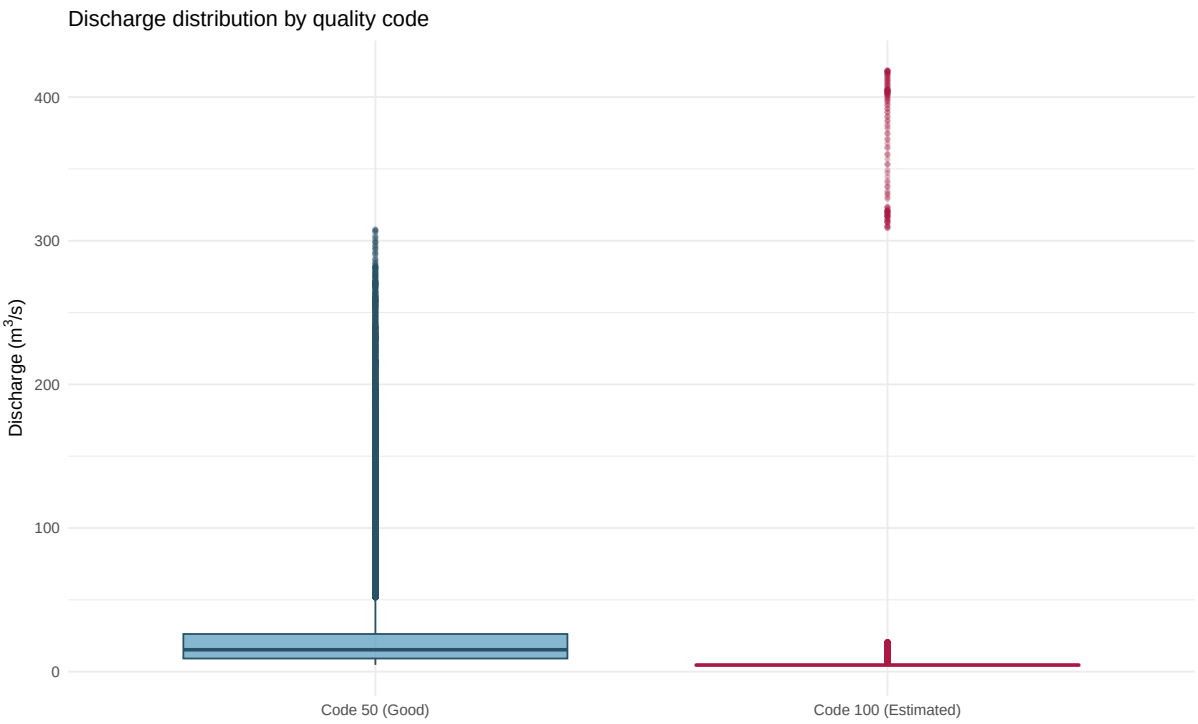


Figure 1: Discharge distribution for quality codes 50 & 100.

2.2 Exploratory Analysis

2.2.1 Hydrograph and Distribution

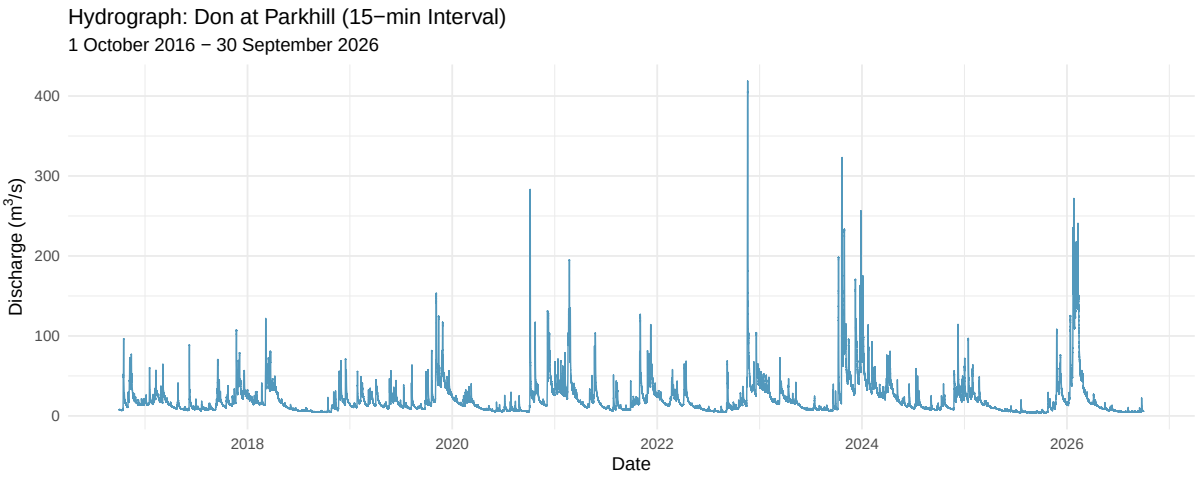
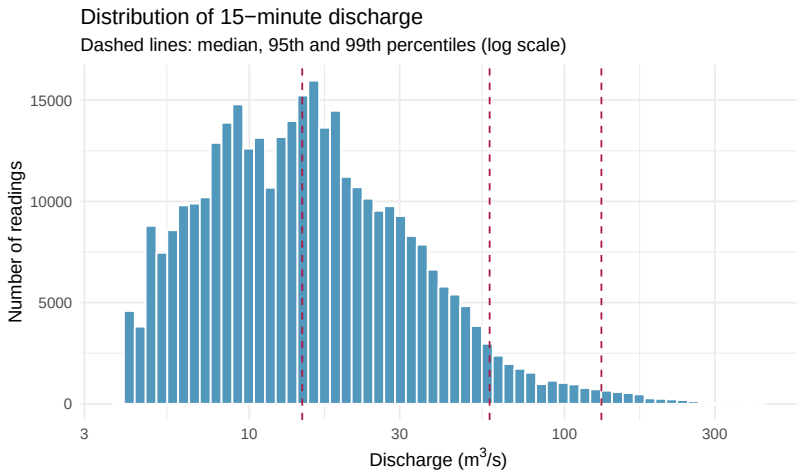


Figure 2: Hydrograph of 15-minute discharge for the Don at Parkhill.



Statistic	Value (m³/s)
Mean	21.65
Standard deviation	24.38
Minimum	3.90
25th percentile	8.74
Median	14.72
75th percentile	25.44
95th percentile	57.88
99th percentile	130.96
Maximum	418.68

Figure 3: Distribution of 15-minute discharge on a log scale, with summary statistics. Dashed lines mark the median, 95th and 99th percentiles.

Discharge distribution is strongly right-skewed (Figure 3), with a median reading of 14.73 m^3/s against a 99th percentile of 130.97 m^3/s and a maximum of 418.68 m^3/s, suggesting extreme flows behave differently from typical flows and need separate analysis.

2.2.2 Seasonality of High Flows

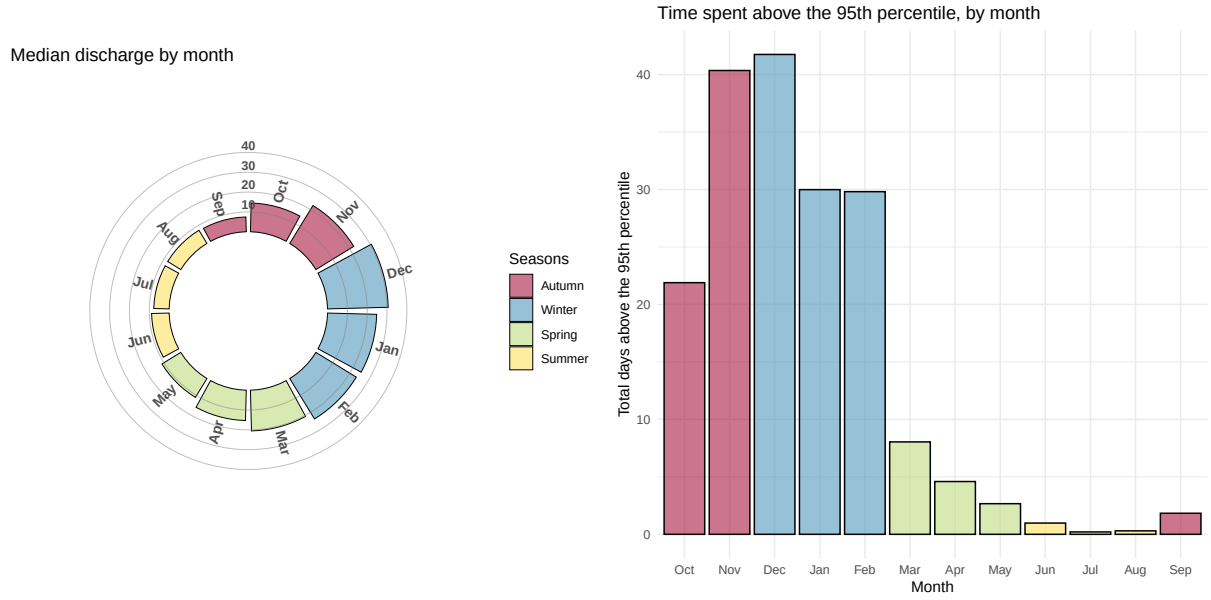


Figure 4: Median discharge by month (left) and total days spent above the 95th percentile in each month (right).

Median discharge is highest from December to February and lowest from July to September, while time spent above the 95th percentile is concentrated even more tightly to between October and February (Figure 4).

2.2.3 Flood Event Hydrographs

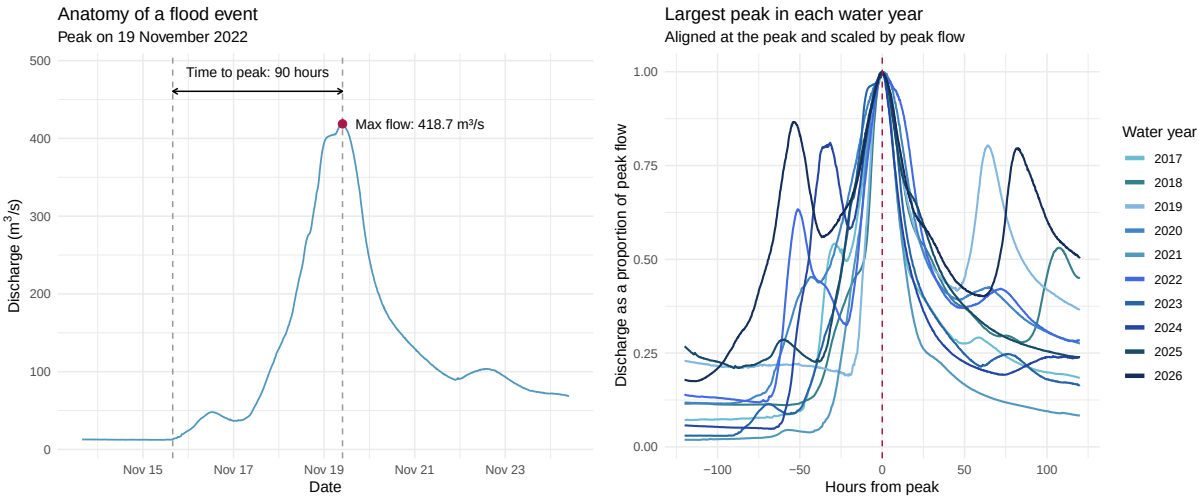


Figure 5: The largest recorded flood event, showing the max flow and time to peak (left), and the largest peak in each water year, aligned at the peak and scaled by peak flow (right).

Figure 5 shows the maximum flow and time to peak of the largest event, and that water-year maxima vary considerably in shape, so treating hydrograph shape as fixed ignores an important source of uncertainty, while the smaller peaks clustering around each maximum mean independent events must be separated.

3 Threshold Selection and Independence

3.1 Robson & Reed (1999) independence rule assumptions

- A *clean peak* is the highest candidate discharge within a 14-day symmetrical window (±7 days), reliably isolating major flood peaks.
- Start of the rise (t_{rise}):

$$t_{\text{rise}} = \max\{t \in [t_{\text{peak}} - 7 \text{ days}, t_{\text{peak}}) \mid q(t) \leq 1.05 \cdot q_{\text{min}}\}$$

  where $q_{\text{min}} = \min_{t \in [t_{\text{peak}} - 7 \text{ days}, t_{\text{peak}})} q(t)$ is the lowest discharge recorded in the 7 days before the peak. Selecting t_{rise} using this condition reliably captures the exact moment discharge begins climbing towards the peak. The 7-day look-back period aligns well with the synoptic cycles of typical flood-generating storms (3-7 days).

3.2 Initial Threshold

We have confirmed the need to extract high flows, and we use Peaks Over Threshold (POT), which takes flood events as peaks of flow above a threshold u, usually chosen as the 95th percentile of flow or the level giving 3 to 5 peaks per year. Figure 6 shows both on the hydrograph: the 95th percentile is 57.9 m^3/s, while counting each exceedance period as a peak gives at least 5, 4 and 3 peaks per year at 79, 93 and 113 m^3/s, though this overstates the number of distinct events, as one flood can cross the threshold several times. The two rules disagree here, as the 95th percentile yields 9.4 peaks per year, so we follow the peaks-per-year rule, which controls the number of events directly rather than the proportion of time above the threshold, and take

$u = 79\ m^3/s$, giving 5 peaks per year, as our initial threshold. We choose the top of the range because Robson and Reed (1999) note this leaves the flexibility to raise it later to exclude smaller floods, and the threshold may need revising once independence is applied.

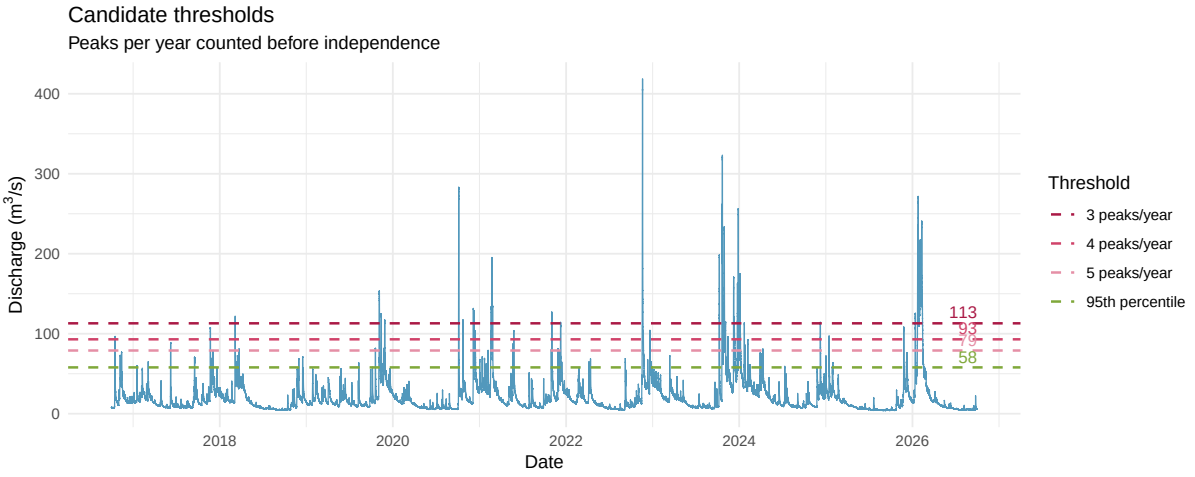


Figure 6: Hydrograph with the candidate thresholds. Peaks per year are counted before applying the independence rules.

3.3 Independence of Peaks

We implement our assumptions to apply the Robson and Reed (1999) independence rules. If either rule fails, we treat the peaks as one event and keep the larger peak.

This reduces the 56 candidate peaks to 35 independent events, or 3.5 per year (Figure 7).

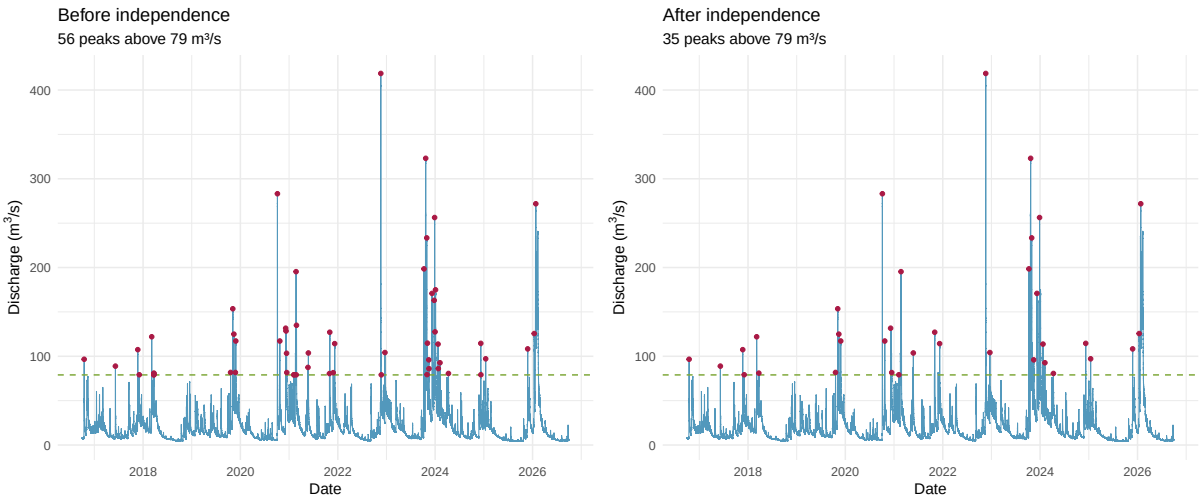


Figure 7: Peaks above the initial threshold before (left) and after (right) applying the independence rules.

3.4 Final Threshold

Since separating events lowers the number of peaks, we recheck the threshold on independent peaks only. The initial threshold gives 3.5 independent peaks per year, while thresholds between 57 and 81 m^3/s give 3 to 5 (Figure 8).

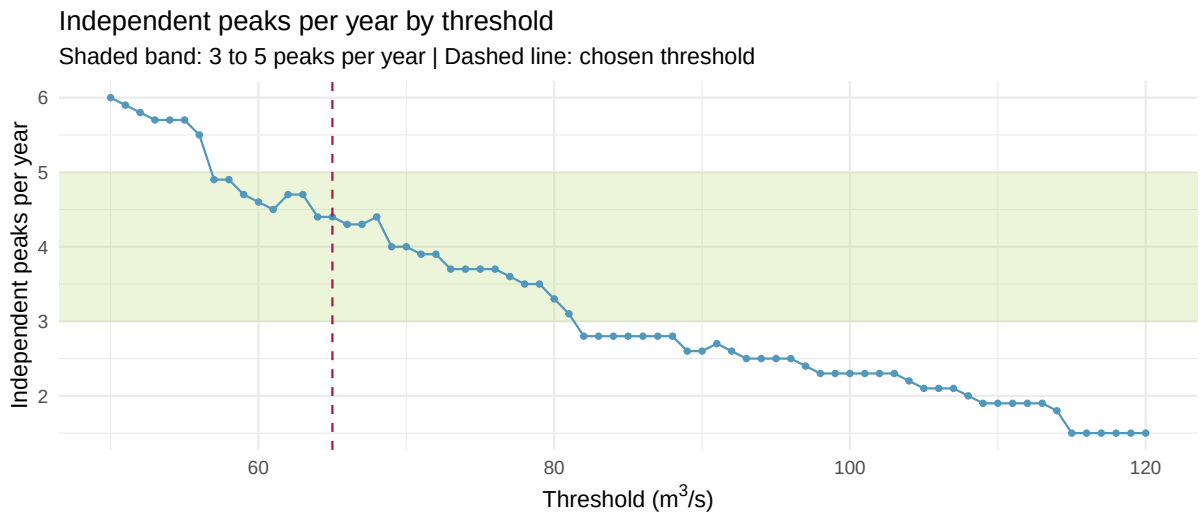


Figure 8: Independent peaks per year at each threshold.

To choose among these, we use the result that excesses over a high enough threshold follow a Generalised Pareto Distribution,

$$P(X - u \leq y \mid X > u) = 1 - \left(1 + \frac{\xi y}{\sigma}\right)^{-1/\xi}, \quad y > 0, \quad 1 + \frac{\xi y}{\sigma} > 0,$$

with scale σ and shape ξ. So that above a valid threshold the mean excess $E[X - u \mid X > u]$ is linear in u and the estimates of ξ and the modified scale $\sigma - \xi u$ stay roughly constant (Coles 2001). From Figure 9, the mean residual life plot is approximately linear above ... m^3/s and the parameter estimates are stable from ... m^3/s.

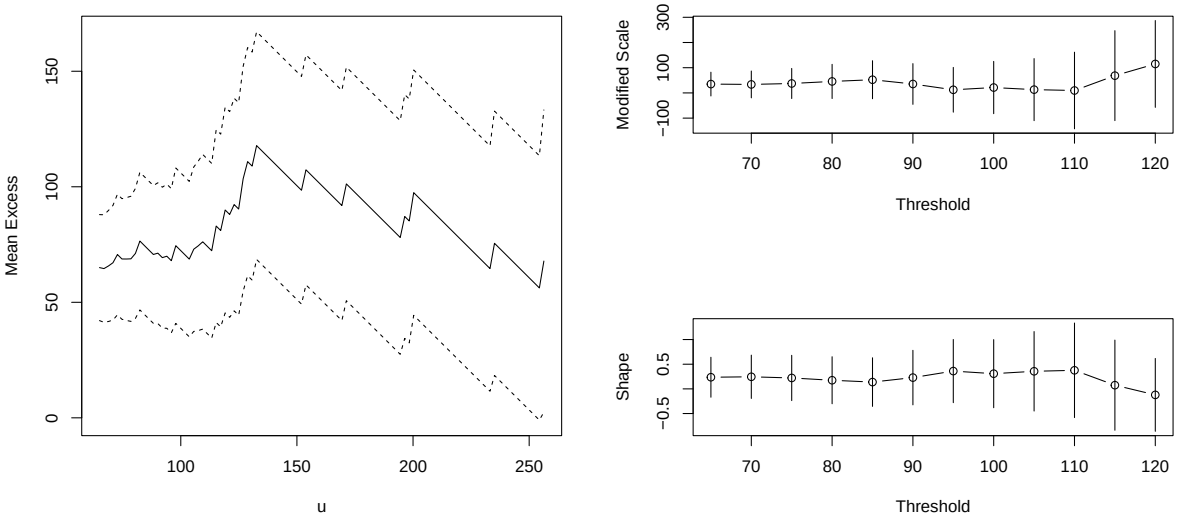


Figure 9: Mean residual life plot (left) and parameter stability plots (right) for the independent peaks at the final threshold.

We therefore lower the threshold to $u = 65$ m^3/s, giving 44 independent events (4.4 per year), with fitted parameters $\hat{\sigma} = 50.34$ (standard error12.76) and $\hat{\xi} = 0.236$ (standard error 0.207). [The diagnostic plots in Figure 10 show ...]

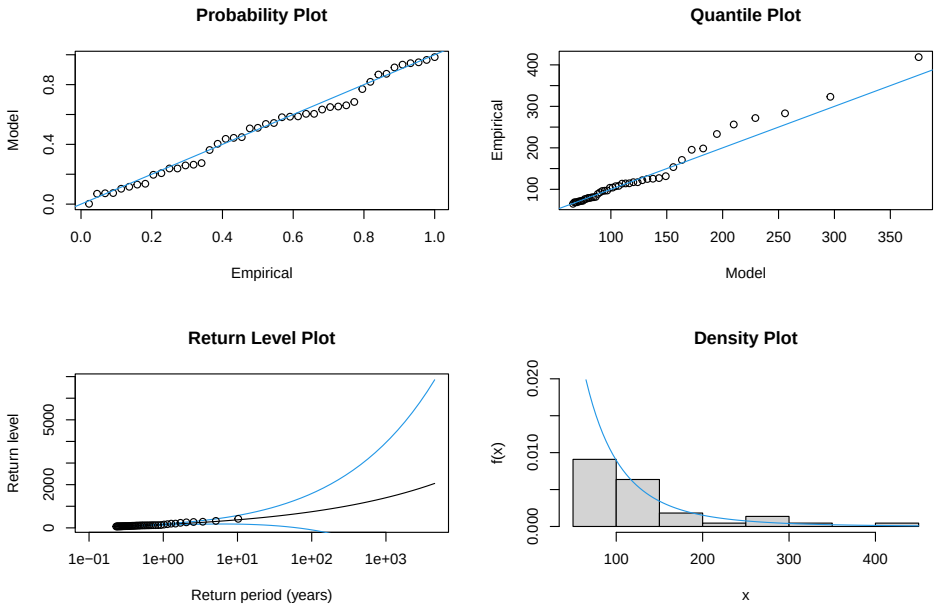


Figure 10: Diagnostic plots for the GPD fitted at the final threshold.

4 Uncertainty Analysis

The final threshold of $u = 65\ m^3/s$ identifies 44 independent flood events over the ten water years, across which maximum flow and time to peak both varied considerably (Figures 11 and 12. Maximum flow had a median of 105.83 m^3/s (interquartile range 79.2–128.2 m^3/s), while time to peak had a median of 44.12 hours (interquartile range 25.9–77.9 hours) and ranged from 0.2 to 151.8 hours. This suggests that extreme flood hydrographs cannot be adequately characterised by a single peak magnitude or a fixed time to peak.

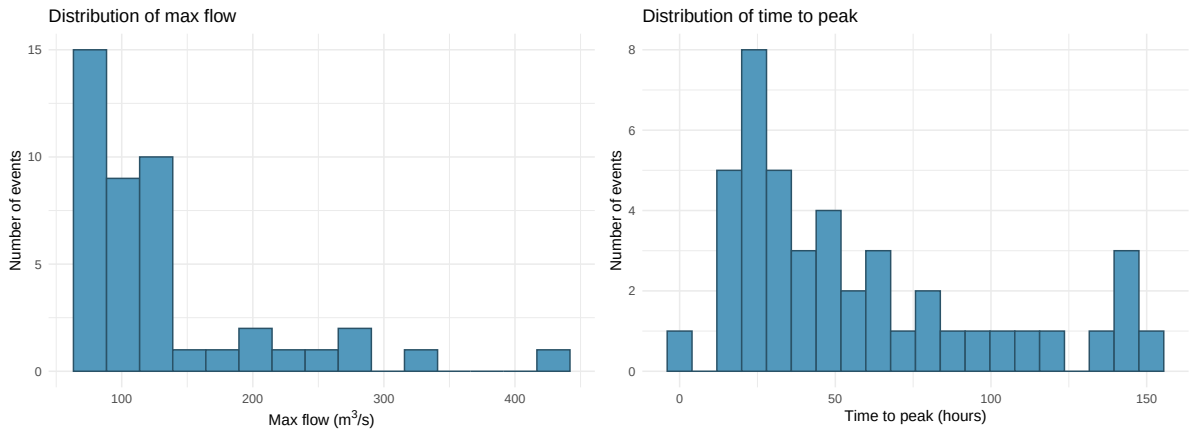


Figure 11: The Distribution of the two parameters

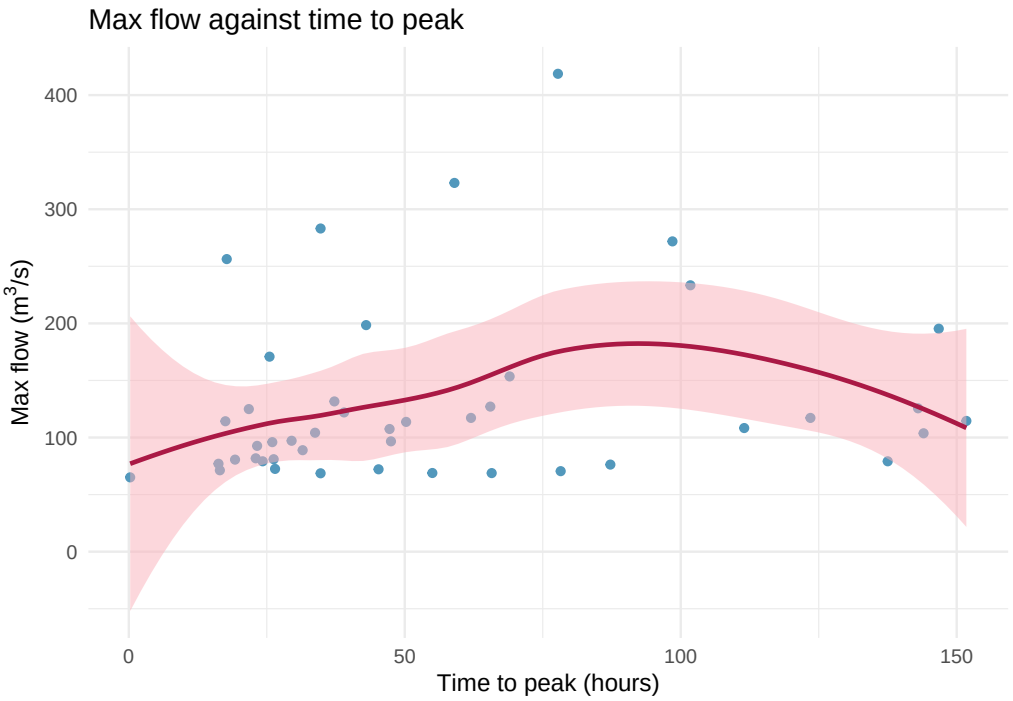


Figure 12: Max flow against time to peak, with a smoothed trend

Since both distributions are skewed and we are interested in any monotonic relationship, we used Spearman's rank correlation, obtaining $\rho = 0.287$ ($p = 0.059$). This indicates a weak positive association, but with insufficient evidence of a monotonic relationship at the 5% significance level, suggesting that larger floods do not necessarily take longer to reach their peak.

Table 2: Spearman's Rank Correlation Between Peak Discharge and Time to Peak

Test	Rho	P-value	S statistic
Spearman's rank correlation	0.287	0.059	10111.57

Table 3: Summary of Max Flow and Time to Peak for each Season

season	Events	Median max flow	Median time to peak (h)
Winter	18	108.951	35.500
Spring	6	80.812	30.625
Summer	1	88.889	31.500
Autumn	19	117.185	59.000

Autumn events (September–November, $n = 19$) had a median maximum flow of 117.19 m^3/s and time to peak of 59 hours, compared with 108.95 m^3/s and 35.5 hours for winter events (December–February, $n = 18$), suggesting that autumn floods tend to take longer to peak (Table 3, Figure 13). With so few events per season, however, this difference should be treated with caution, and spring (6) and summer (1) had too few events to compare.

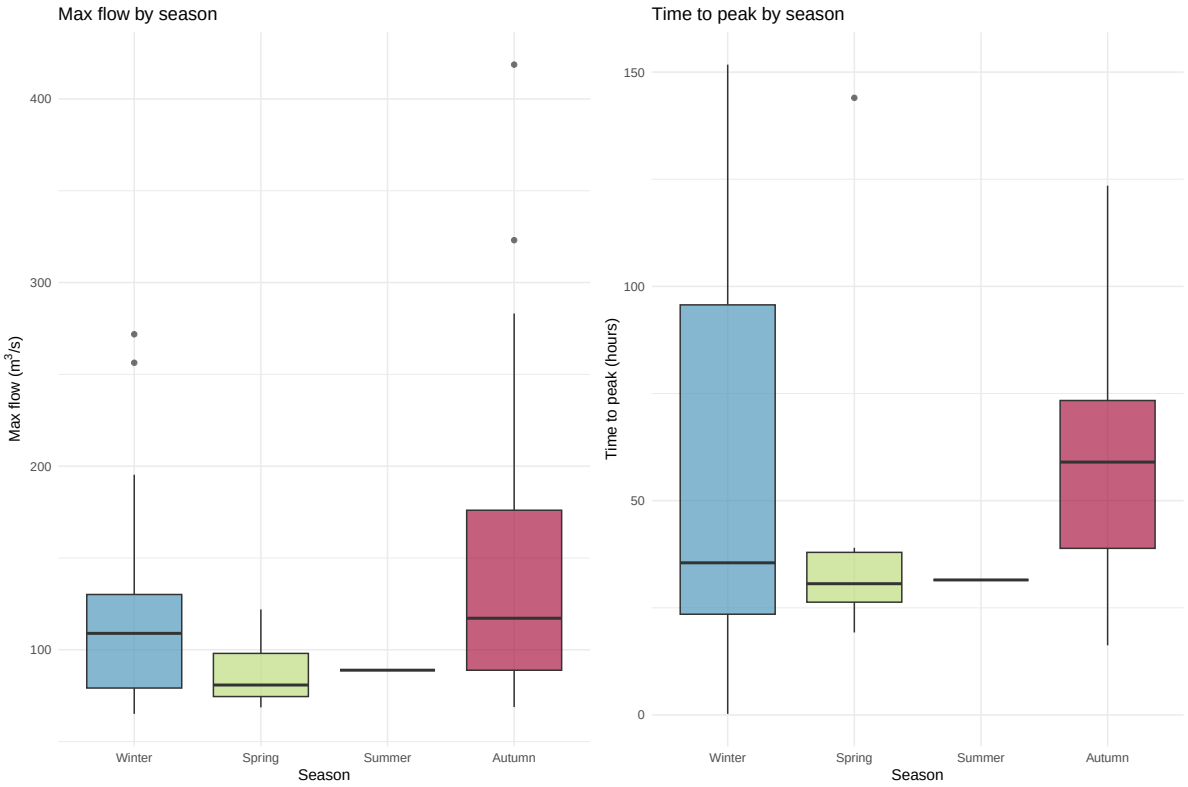


Figure 13: A comparison of the uncertainty in the parameters by season

Table 4: Final Uncertanity Table

threshold	events	Events per Year	Median Max Flow	Median Time to Peak (Hours)	Spearman Rho Corelation
55	57	5.7	81.792	43.000	0.066
60	46	4.6	100.434	44.125	0.184
65	44	4.4	105.834	44.125	0.287
70	40	4.0	111.001	44.125	0.288
75	37	3.7	114.256	47.250	0.220

Repeating the analysis at thresholds of 60-75 m^3/s, median time to peak remained between 43 and 47.25 hours and Spearman's correlation between 0.184 and 0.287 (Table 4, suggesting that the observed variability in time to peak is not simply a consequence of the chosen threshold.

References

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